

Day 11: AI Cybersecurity Assistant (Chatbot) & Email Threat Simulator

Key Milestone: Interactive AI Assistant & Forensic Sandbox Simulator

Sprint Focus: Execution of core deliverables for Day 11 spanning AI/NLP Intent Classification, Email Header Forensics, IOC Intelligence, Relationship Graphing, Case Workflow, and Full-Stack Frontend.

Key Deliverables Completed: AI Security Chatbot Module, Threat Simulator Sandbox, Interactive Playbook Engine.

Individual Team Member Task Logs & Technical Contributions

- Muniswami.Y — AI / NLP Lead

Domain Ownership: Explainable AI, Intent & Phishing Detection, Social Engineering signals | **Key Modules:** NLP/AI Engine, Explainability, Content Risk scoring

Day 11 Execution: Integrated generative AI chat completions with custom cybersecurity system prompts grounded strictly in active case telemetry.
- Jeevan reddy.G — Header & Forensics Lead

Domain Ownership: Email header analysis, SPF/DKIM/DMARC, Mail path reconstruction, Timestamp analysis | **Key Modules:** Header Forensics, Mail Path, Infrastructure Risk

Day 11 Execution: Built simulated attack generators (CEO Fraud, Fake Invoice, Ransomware, Credential Harvesting) with randomized header topologies.
- Lahari.K — IOC & Infrastructure Lead

Domain Ownership: URL, Domain, IP extraction, Geolocation, ASN/ISP mapping, Malicious indicator detection | **Key Modules:** IOC Extraction, Geolocation, URL/Domain/IP Risk

Day 11 Execution: Created synthetic IOC generator to safely simulate malicious domains, fast-flux IPs, and lookalike brand infrastructures.
- Hemanth.P — Graph & Campaign Intelligence Lead

Domain Ownership: Attack Relationship Graph, Similarity detection, Campaign grouping & correlation | **Key Modules:** Graph Engine, Campaign Detection, Similar Attack Detection

Day 11 Execution: Implemented real-time graph node highlighting triggered by user queries in the AI Assistant chatbot.
- Abbas.N — Risk, Case & Response Lead

Domain Ownership: Dynamic Threat Score engine, Automatic Forensic Case creation, Recommended Response actions | **Key Modules:** Risk Scoring, Case Management, Response Recommendations

Day 11 Execution: Created interactive mitigation playbook engine providing step-by-step containment checklists based on incident severity.
- Moulaanbee.N — Frontend + Assistant + Simulator Lead

Domain Ownership: Dashboard, Visualizations (Graph, Map, Timeline), AI Chatbot, Email Threat Simulator, Overall UI/UX | **Key Modules:** Dashboard, AI Cybersecurity Assistant, Threat Simulator, Frontend Integration

Day 11 Execution: Built the full-screen Threat Simulator sandbox and collapsible floating AI Cybersecurity Assistant chat drawer.

System Integration & Quality Verification

Cross-Module Integration: Tested AI Chatbot querying live case context: verified zero hallucinations and exact grounding in active header evidence.

Quality Assurance & Verification: All technical modules, automated unit test suites, and API contracts for Day 11 executed with 100% clean passes. Code reviewed and merged into main repository.

Prepared By: Threat Trace AI Development Team | **Project Lead:** Muniswami.Y | **Status:** SPRINT DAY 11 MILESTONE SIGNED OFF